

Midterm Review

Preliminary Notes

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Abstract

We study a risk-neutral principal who may review Gaussian cumulative output at an endogenous date and condition bounded, limited-liability compensation and continuation on performance. The optimal review menu has two, three, or four ordered regions: a bad stop, low-rent continuation, high-rent continuation, and possibly a good stop with the maximal prize. Review timing balances the informativeness of output against continuation rents available as incentive capital. Arbitrarily early reviews reduce value, whereas sufficiently tight feasible payment caps make an interior review strictly optimal. The model identifies purely incentive-driven termination and rent recycling in a canonical net-output agency problem.

1 Introduction

Interim performance reviews are commonly used to decide whether an agent should be retained, how demanding future targets should be, and whether exceptional early performance should be rewarded immediately. These decisions are usually analyzed separately. This note studies them jointly in a canonical moral-hazard environment. A risk-neutral principal receives terminal output net of compensation, a risk-neutral agent chooses phase-level binary effort, cumulative output follows a Brownian motion, and compensation is subject to limited liability and a hard upper bound. The principal may commit to a single public review at an endogenous date τ and condition the post-review opportunity on the realized state.

Without a review, bounded payments lead to the familiar likelihood-ratio bonus: the agent receives B only when terminal output exceeds a threshold. The dynamic problem becomes substantially richer because continuation itself creates rents. Those rents need not be treated simply as an agency loss. By assigning more valuable continuation opportunities to better review outcomes, the principal recycles unavoidable future rents into incentives for first-phase effort. The review therefore allocates what may be called *incentive capital*.

The first main result is an ordered performance ladder. For a fixed review date, the post-review full-effort frontier is a line segment between a minimum-rent contract and a maximum-rent contract. Strict MLRP at the review and linearity of the principal's assignment problem imply that only the endpoints of this frontier and the endpoints of the no-effort frontier are used. Depending on parameters, the optimal review scheme has two, three, or four nonempty regions. Poor outcomes receive a *bad stop*, termination, with zero payment. Adequate outcomes receive low-rent continuation under a demanding terminal bonus. Better outcomes receive high-rent continuation under an easier bonus. When first-phase incentives are especially costly, exceptional outcomes receive a good stop: the principal pays B immediately at the time of the review and sacrifices second-phase production. Because the productivity of future effort is independent of the review state, both stopping regions are purely incentive driven rather than responses to bad or good project types.

The second main result concerns review timing. Waiting longer separates the review-state distributions under work and shirking and therefore improves the statistical quality of performance measurement. But delay also shortens the continuation phase, reducing both its productive surplus and the rents that can be awarded as prizes for first-phase effort. The optimal review date τ balances information against incentive capital. This is a genuinely dynamic tradeoff: the principal chooses not only how to reward performance, but also how much future agency rent to create for use as a current incentive instrument.

The endpoint behavior makes this tradeoff especially sharp. The right derivative of the review value at $\tau = 0$ is minus infinity. An arbitrarily short probationary phase is therefore harmful for incentive provision, and every strictly optimal review is bounded away from the initial date. Short probationary periods observed in practice must be justified in this model by motives outside pure moral hazard, such as screening, match learning, or technological experimentation. At the opposite endpoint, the left derivative at $\tau = T$ is explicit. When the payment cap is sufficiently tight, carving out even a short continuation phase has first-order value because it creates scarce continuation rents. If the no-review full-effort benchmark is nonnegative at the feasibility boundary, every sufficiently tight feasible cap makes an interior review strictly optimal. With a loose cap, by contrast, the terminal contract can provide incentives directly and the marginal value of a continuation phase disappears.

The analysis proceeds as follows. Section 2 derives the no-review benchmark, and Section 3 constructs the post-review continuation frontier. Sections 4 and 5 formulate pre-review incentives and characterize the ordered menu and the boundaries between its two-, three-, and four-region forms. Section 6 derives the review value, endpoint derivatives, and the tight-cap result. Section 7 relates these mechanisms to work on bounded payments, interim evaluation, termination, and feedback design.

2 Environment and the no-review benchmark

There is a risk-neutral principal and a risk-neutral agent. The project runs over the interval $[0, T]$, and its state evolves according to

$$dX_t = e_t dt + dW_t, \quad X_0 = 0, \quad (1)$$

where W is a standard Brownian motion.

Effort is chosen at the phase level. In any phase of length t , the agent chooses either to work throughout the phase or to shirk throughout the phase:

$$e \in \{0, 1\}. \quad (2)$$

Working throughout a phase of length t costs the agent ct , where

$$c \in (0, 1). \quad (3)$$

The same flow-cost parameter c applies before and after any review. This restriction is important because the review date will eventually be chosen endogenously.

The principal receives the terminal state net of payments to the agent. The agent is protected by limited liability, and total compensation is bounded above by B :

$$0 \leq w \leq B. \quad (4)$$

Both parties observe the state only at contractually specified review dates and at the terminal date. In this section there is no review, so the only public signal is X_T .

We assume throughout that c is sufficiently low and B is sufficiently large that full effort is feasible over every phase length that arises in the analysis. In particular, full effort is feasible over the entire horizon $[0, T]$.

2.1 The no-review contracting problem

Without a review, the agent makes a single phase-level effort choice

$$e \in \{0, 1\}. \quad (5)$$

If the agent works, then

$$X_T \mid e = 1 \sim N(T, T), \quad (6)$$

whereas if he shirks,

$$X_T \mid e = 0 \sim N(0, T). \quad (7)$$

Let f_1 and f_0 denote the corresponding densities. A terminal contract is a measurable function

$$w : \mathbb{R} \longrightarrow [0, B]. \quad (8)$$

The agent works if and only if

$$\int_{\mathbb{R}} w(x) [f_1(x) - f_0(x)] dx \geq cT. \quad (9)$$

The principal's problem conditional on inducing work is therefore

$$\min_{0 \leq w \leq B} \int_{\mathbb{R}} w(x) f_1(x) dx \quad (10)$$

subject to (9).

The likelihood ratio is

$$\frac{f_1(x)}{f_0(x)} = \exp\left(x - \frac{T}{2}\right), \quad (11)$$

which is strictly increasing in x . Equivalently,

$$1 - \frac{f_0(x)}{f_1(x)} = 1 - \exp\left(\frac{T}{2} - x\right) \quad (12)$$

is strictly increasing in x . Thus high terminal states provide the most incentive per dollar of expected compensation under work.

A Neyman–Pearson argument implies that the cheapest contract inducing work pays the maximal amount on an upper tail of the terminal-state distribution:

$$w^{\text{NR}}(x) = B \mathbf{1}\{x \geq k_T\}. \quad (13)$$

To characterize the threshold, define

$$D(t, z) = \Phi(z) - \Phi(z - \sqrt{t}), \quad (14)$$

where Φ is the standard normal distribution function. Also define

$$z_T = \frac{T - k_T}{\sqrt{T}}. \quad (15)$$

Under work, the probability of receiving the bonus is

$$\mathbb{P}\{X_T \geq k_T \mid e = 1\} = \Phi(z_T), \quad (16)$$

while under shirking it is

$$\mathbb{P}\{X_T \geq k_T \mid e = 0\} = \Phi(z_T - \sqrt{T}). \quad (17)$$

The incentive constraint is therefore

$$BD(T, z_T) \geq cT. \quad (18)$$

The function $D(T, z)$ is strictly increasing for

$$z < \frac{\sqrt{T}}{2}, \quad (19)$$

strictly decreasing for

$$z > \frac{\sqrt{T}}{2}, \quad (20)$$

and attains its maximum at

$$z = \frac{\sqrt{T}}{2}. \quad (21)$$

Its maximum value is

$$\bar{D}(T) = 2\Phi\left(\frac{\sqrt{T}}{2}\right) - 1. \quad (22)$$

Full effort is feasible if and only if

$$B\left[2\Phi\left(\frac{\sqrt{T}}{2}\right) - 1\right] \geq cT. \quad (23)$$

Suppose this inequality is strict. Then the binding equation

$$BD(T, z) = cT \quad (24)$$

has two solutions. Let z_T^L denote the smaller solution:

$$z_T^L < \frac{\sqrt{T}}{2}. \quad (25)$$

Because expected compensation under work,

$$B\Phi(z), \quad (26)$$

is increasing in z , the principal chooses the smaller root. The optimal no-review threshold is therefore

$$\boxed{k_T^{\text{NR}} = T - z_T^L \sqrt{T}}, \quad (27)$$

where z_T^L is the smaller solution to

$$\boxed{B\left[\Phi(z_T^L) - \Phi(z_T^L - \sqrt{T})\right] = cT}. \quad (28)$$

The optimal no-review contract is consequently

$$\boxed{w^{\text{NR}}(X_T) = B\mathbf{1}\left\{X_T \geq T - z_T^L \sqrt{T}\right\}}. \quad (29)$$

2.2 Expected compensation and agency rent

Expected compensation under work is

$$p^{\text{NR}} = B\Phi(z_T^L). \quad (30)$$

Using the binding incentive constraint,

$$B\Phi(z_T^L) = cT + B\Phi(z_T^L - \sqrt{T}). \quad (31)$$

Thus the agent's expected utility under work is

$$R^{\text{NR}} = p^{\text{NR}} - cT \quad (32)$$

$$= B\Phi(z_T^L - \sqrt{T}). \quad (33)$$

This rent has a simple interpretation. It is exactly the expected payment the agent would receive if he shirked:

$$R^{\text{NR}} = \mathbb{E} [w^{\text{NR}}(X_T) \mid e = 0]. \quad (34)$$

Limited liability prevents the principal from charging the agent for favorable Brownian realizations that occur even under shirking. The agent must therefore receive a strictly positive rent whenever full effort is induced by a nontrivial bounded contract.

The contract in (29) makes the agent indifferent between work and shirking. Under the usual weak-obedience convention, the agent selects work. If strict incentives are required, the principal may lower the threshold slightly, increasing expected compensation by an arbitrarily small amount.

2.3 The principal's no-review value

Conditional on work, expected terminal output is

$$\mathbb{E}[X_T \mid e = 1] = T. \quad (35)$$

The principal's expected payoff is therefore

$$\boxed{V^{\text{NR}}(T; c, B) = T - B\Phi(z_T^L)}. \quad (36)$$

Equivalently, using the agent's rent,

$$V^{\text{NR}}(T; c, B) = (1 - c)T - R^{\text{NR}}, \quad (37)$$

or

$$\boxed{V^{\text{NR}}(T; c, B) = (1 - c)T - B\Phi(z_T^L - \sqrt{T})}. \quad (38)$$

The first-best surplus is

$$V^{\text{FB}}(T) = (1 - c)T. \quad (39)$$

Hence the entire agency loss in the no-review benchmark is the limited-liability rent:

$$\boxed{V^{\text{FB}}(T) - V^{\text{NR}}(T; c, B) = R^{\text{NR}} = B\Phi(z_T^L - \sqrt{T})}. \quad (40)$$

The principal can always offer the zero contract, induce shirking, and obtain expected payoff zero. Her full no-review value is therefore

$$\boxed{\bar{V}^{\text{NR}}(T; c, B) = \max \{0, T - B\Phi(z_T^L)\}}. \quad (41)$$

In the parameter region of primary interest, full effort is both feasible and profitable, so

$$\bar{V}^{\text{NR}}(T; c, B) = V^{\text{NR}}(T; c, B). \quad (42)$$

This no-review value will be the benchmark against which the endogenous-review contract is evaluated.

3 The post-review continuation frontier

Fix a review date $\tau \in (0, T)$ and let

$$h = T - \tau \quad (43)$$

be the length of the continuation phase. At the review, the state $X_\tau = x$ is publicly observed. Because the technology is translation invariant, a continuation contract can be written as a function of the terminal increment

$$Y = X_T - X_\tau. \quad (44)$$

If the agent works throughout the continuation phase, then

$$Y \mid e_2 = 1 \sim N(h, h), \quad (45)$$

whereas if he shirks throughout the continuation phase,

$$Y \mid e_2 = 0 \sim N(0, h). \quad (46)$$

Let f_1^h and f_0^h denote the corresponding densities.

A continuation contract is a measurable function

$$w : \mathbb{R} \longrightarrow [0, B]. \quad (47)$$

The agent works throughout the continuation phase if and only if

$$\int_{\mathbb{R}} w(y) [f_1^h(y) - f_0^h(y)] dy \geq ch. \quad (48)$$

If work is induced, define the agent's continuation rent by

$$r(w; h) = \int_{\mathbb{R}} w(y) f_1^h(y) dy - ch. \quad (49)$$

The principal's incremental continuation payoff, excluding the already realized state x , is

$$s(w; h) = h - \int_{\mathbb{R}} w(y) f_1^h(y) dy \quad (50)$$

$$= (1 - c)h - r(w; h). \quad (51)$$

Thus every full-effort continuation contract lies on the line

$$s + r = (1 - c)h. \quad (52)$$

The remaining question is which continuation rents can be generated by contracts satisfying limited liability, the payment cap, and the incentive constraint.

3.1 Feasibility of full continuation effort

For an upper-tail bonus with threshold K , write

$$w_K(y) = B\mathbf{1}\{y \geq K\} \quad (53)$$

and define

$$z = \frac{h - K}{\sqrt{h}}. \quad (54)$$

Under work, the probability of receiving the bonus is

$$\Phi(z), \quad (55)$$

while under shirking it is

$$\Phi(z - \sqrt{h}). \quad (56)$$

Hence the incentive difference generated by the contract is

$$BD(h, z) = B \left[\Phi(z) - \Phi(z - \sqrt{h}) \right]. \quad (57)$$

The function $D(h, z)$ is maximized at

$$z = \frac{\sqrt{h}}{2}, \quad (58)$$

with maximum

$$\overline{D}(h) = 2\Phi\left(\frac{\sqrt{h}}{2}\right) - 1. \quad (59)$$

It follows that full continuation effort is feasible if and only if

$$B\overline{D}(h) = B \left[2\Phi\left(\frac{\sqrt{h}}{2}\right) - 1 \right] \geq ch. \quad (60)$$

We maintain the stronger assumption that this inequality is strict for every continuation horizon that arises below.

Under strict feasibility, the equation

$$BD(h, z) = ch \quad (61)$$

has exactly two solutions. Denote them by

$$z_L(h) < \frac{\sqrt{h}}{2} < z_H(h). \quad (62)$$

The corresponding terminal thresholds are

$$K_j(h) = h - z_j(h)\sqrt{h}, \quad j \in \{L, H\}. \quad (63)$$

Since $z_L(h) < z_H(h)$, the low-rent contract has the harder performance threshold:

$$K_L(h) > K_H(h). \quad (64)$$

3.2 The minimum- and maximum-rent contracts

Proposition 1 (The full-effort continuation frontier). *Fix $h > 0$ and suppose that*

$$B \left[2\Phi \left(\frac{\sqrt{h}}{2} \right) - 1 \right] > ch. \quad (65)$$

Then the following statements hold.

1. *The full-effort contract that minimizes the agent's continuation rent is*

$$w_L^h(y) = B \mathbf{1}\{y \geq K_L(h)\}. \quad (66)$$

2. *The full-effort contract that maximizes the agent's continuation rent is*

$$w_H^h(y) = B \mathbf{1}\{y \geq K_H(h)\}. \quad (67)$$

3. *The corresponding rents are*

$$r_j(h) = B\Phi(z_j(h)) - ch \quad (68)$$

$$= B\Phi(z_j(h) - \sqrt{h}), \quad j \in \{L, H\}, \quad (69)$$

and satisfy

$$0 < r_L(h) < r_H(h) < B - ch. \quad (70)$$

4. *The set of continuation rents attainable while inducing full effort is exactly*

$$\mathcal{R}_h^B = [r_L(h), r_H(h)]. \quad (71)$$

Consequently, the principal's full-effort continuation frontier is

$$\boxed{S_h^B(r) = (1 - c)h - r, \quad r \in [r_L(h), r_H(h)]}. \quad (72)$$

Proof. Consider first the problem of minimizing expected compensation under work:

$$\min_{0 \leq w \leq B} \int_{\mathbb{R}} w(y) f_1^h(y) dy \quad (73)$$

subject to (48). The incentive generated per dollar of expected compensation under work is

$$\frac{f_1^h(y) - f_0^h(y)}{f_1^h(y)} = 1 - \frac{f_0^h(y)}{f_1^h(y)} = 1 - \exp\left(\frac{h}{2} - y\right), \quad (74)$$

which is strictly increasing in y . A Neyman–Pearson argument therefore implies that the cheapest way to generate a given incentive difference is to pay B on an upper tail and zero elsewhere. The incentive constraint binds at the optimum. Of the two upper-tail contracts satisfying (61), expected compensation under work is

$$B\Phi(z), \quad (75)$$

which is increasing in z . Hence the minimizing contract uses the smaller root $z_L(h)$ and is given by (66).

To maximize expected compensation under work while preserving full-effort incentives, define the withheld payment

$$q(y) = B - w(y). \quad (76)$$

Then $0 \leq q \leq B$, and the incentive constraint becomes

$$\int_{\mathbb{R}} q(y) [f_0^h(y) - f_1^h(y)] dy \geq ch. \quad (77)$$

Moreover,

$$\int_{\mathbb{R}} w(y) f_1^h(y) dy = B - \int_{\mathbb{R}} q(y) f_1^h(y) dy. \quad (78)$$

Maximizing expected compensation is therefore equivalent to minimizing expected withheld payment under work subject to (77). The incentive generated per dollar of withheld payment is

$$\frac{f_0^h(y) - f_1^h(y)}{f_1^h(y)} = \exp\left(\frac{h}{2} - y\right) - 1, \quad (79)$$

which is strictly decreasing in y . The cheapest withheld-payment schedule therefore sets $q = B$ on a lower tail and zero elsewhere. Equivalently, w is an upper-tail bonus. Of the two binding upper-tail contracts, the one with the larger root $z_H(h)$ has the larger expected payment under work. This proves (67).

For either extreme contract, the binding incentive constraint gives

$$B\Phi(z_j(h)) - B\Phi(z_j(h) - \sqrt{h}) = ch. \quad (80)$$

Thus

$$r_j(h) = B\Phi(z_j(h)) - ch = B\Phi(z_j(h) - \sqrt{h}). \quad (81)$$

Since $z_L(h) < z_H(h)$ and Φ is strictly increasing, the rent ordering in (70) follows.

Finally, the first two parts establish that no full-effort contract can generate rent below $r_L(h)$ or above $r_H(h)$. Conversely, for every $\alpha \in [0, 1]$, the contract

$$w_\alpha^h = (1 - \alpha)w_L^h + \alpha w_H^h \quad (82)$$

satisfies $0 \leq w_\alpha^h \leq B$, induces full effort with a binding incentive constraint, and generates rent

$$r_\alpha(h) = (1 - \alpha)r_L(h) + \alpha r_H(h). \quad (83)$$

Hence every rent in $[r_L(h), r_H(h)]$ is attainable. Equation (51) then yields the frontier in (72). $\square$

Remark 1 (Hard and easy continuation contracts). *The minimum-rent contract w_L^h sets a relatively high terminal threshold and therefore gives the agent a small probability of receiving B even when he works. The maximum-rent contract w_H^h sets a lower threshold and gives the agent a much larger probability of receiving B . Both contracts generate exactly the same work-versus-shirk incentive difference, ch , but they differ in the amount of rent paid in states that are also likely under shirking. These two extreme rents will become the building blocks of the optimal review menu.*

3.3 The no-effort frontier

For later use, it is also useful to record the efficient contracts that induce no continuation effort. If the agent shirks in the continuation phase and receives expected utility $r \in [0, B]$, the principal's incremental payoff is

$$-r. \quad (84)$$

This payoff is implemented by the constant contract

$$w(y) = r. \quad (85)$$

Thus the no-effort frontier is

$$\boxed{S_{h,0}^B(r) = -r, \quad r \in [0, B].} \quad (86)$$

Its endpoints are termination with zero payment,

$$(r, s) = (0, 0), \quad (87)$$

and a maximal cash prize with no continuation effort,

$$(r, s) = (B, -B). \quad (88)$$

The next section will show how the principal assigns these no-effort and full-effort continuation opportunities to different review states in order to induce first-phase effort.

4 Stopping at the review and pre-review obedience

Fix a review date $\tau \in (0, T)$ and continue to write

$$h = T - \tau \quad (89)$$

for the length of the second phase. This section completes the set of actions available at the review and formulates the principal's problem of inducing work throughout the first phase.

4.1 The review signal

If the agent works throughout the first phase, then

$$X_\tau \mid e_1 = 1 \sim N(\tau, \tau), \quad (90)$$

whereas if he shirks throughout the first phase,

$$X_\tau \mid e_1 = 0 \sim N(0, \tau). \quad (91)$$

Let g_1^τ and g_0^τ denote the corresponding densities.

Lemma 1 (Strict MLRP at the review). *The review signal satisfies strict monotone likelihood ratio property. In particular,*

$$\frac{g_1^\tau(x)}{g_0^\tau(x)} = \exp\left(x - \frac{\tau}{2}\right), \quad (92)$$

and the likelihood ratio is strictly increasing in x .

Proof. Using the two normal densities in (90) and (91),

$$\log \frac{g_1^\tau(x)}{g_0^\tau(x)} = -\frac{(x - \tau)^2}{2\tau} + \frac{x^2}{2\tau} \quad (93)$$

$$= x - \frac{\tau}{2}. \quad (94)$$

Exponentiating gives (92). Strict monotonicity follows immediately. $\square$

4.2 Bad stops, good stops, and continuation actions

At a review state x , describe an action by a pair (r, s) , where r is the agent's continuation utility and s is the principal's incremental payoff from the second phase, excluding the already realized state x .

If the principal does not induce second-phase effort, she may promise any payment $r \in [0, B]$. Literal termination at the review and continuation with zero effort are equivalent in expected surplus: in either case the expected future increment in the state is zero. The no-effort frontier is therefore

$$\mathcal{A}_0(h) = \{(r, -r) : r \in [0, B]\}. \quad (95)$$

Its two endpoints are

$$A_0 = (0, 0), \quad (96)$$

which we call the *bad stop*, and

$$A_B = (B, -B), \quad (97)$$

which we call the *good stop*. The good stop awards the maximal payment after exceptionally strong first-phase performance but induces no second-phase effort.

If the principal induces full second-phase effort, Proposition 1 shows that the feasible utility–surplus pairs form the line segment

$$\mathcal{A}_1(h) = \{(r, (1-c)h - r) : r \in [r_L(h), r_H(h)]\}. \quad (98)$$

Its endpoints are the low-rent continuation action

$$A_L = (r_L(h), (1-c)h - r_L(h)) \quad (99)$$

and the high-rent continuation action

$$A_H = (r_H(h), (1-c)h - r_H(h)). \quad (100)$$

The complete review-action set is

$$\mathcal{A}(h) = \mathcal{A}_0(h) \cup \mathcal{A}_1(h). \quad (101)$$

Lemma 2 (Endpoint reduction). *For any scalar $q \in \mathbb{R}$, a maximizer of the linear objective*

$$s + qr \quad (102)$$

over $(r, s) \in \mathcal{A}(h)$ can be chosen from

$$\{A_0, A_L, A_H, A_B\}. \quad (103)$$

Proof. The objective in (102) is linear in (r, s) . Its maximum over each of the two line segments $\mathcal{A}_0(h)$ and $\mathcal{A}_1(h)$ is therefore attained at an endpoint of that segment. The endpoints are exactly the four actions listed in (103). $\square$

Lemma 2 does not yet determine which action is assigned to each review state. It establishes that, once the first-phase incentive constraint is incorporated through a multiplier, no interior point of either frontier is needed except in knife-edge cases of indifference.

4.3 Pre-review obedience

Let

$$a(x) \in \mathcal{A}(h) \quad (104)$$

be the action assigned after review state x , and write

$$a(x) = (r(x), s(x)). \quad (105)$$

The agent's expected utility from working throughout the first phase is

$$\int_{\mathbb{R}} r(x) g_1^\tau(x) dx - c\tau, \quad (106)$$

whereas his expected utility from shirking throughout the first phase is

$$\int_{\mathbb{R}} r(x) g_0^\tau(x) dx. \quad (107)$$

Thus pre-review obedience requires

$$\boxed{\int_{\mathbb{R}} r(x) [g_1^\tau(x) - g_0^\tau(x)] dx \geq c\tau.} \quad (108)$$

Because $r(x) \geq 0$, the utility from shirking is nonnegative. Hence (108) also guarantees participation when the agent follows the recommended first-phase action.

Conditional on inducing first-phase work, expected output accumulated by the review is τ . The principal's fixed-review-date problem is therefore

$$\boxed{\begin{aligned} V^R(\tau) = \sup_{a(\cdot)} \quad & \tau + \int_{\mathbb{R}} s(x) g_1^\tau(x) dx \\ \text{subject to} \quad & \int_{\mathbb{R}} r(x) [g_1^\tau(x) - g_0^\tau(x)] dx \geq c\tau, \\ & (r(x), s(x)) \in \mathcal{A}(h) \quad \text{for almost every } x. \end{aligned}} \quad (109)$$

The value in (109) is conditional on inducing work in the first phase. The principal can also induce shirking or dispense with the review, alternatives that will be included when the review date is optimized.

4.4 The Lagrangian for pre-review obedience

Let $\lambda \geq 0$ be the multiplier on (108). The Lagrangian for the fixed- τ problem is

$$\mathcal{L}(a; \lambda, \tau) = \tau - \lambda c\tau \quad (110)$$

$$+ \int_{\mathbb{R}} \{s(x) g_1^\tau(x) + \lambda r(x) [g_1^\tau(x) - g_0^\tau(x)]\} dx. \quad (111)$$

Factoring out $g_1^\tau(x)$ gives

$$\mathcal{L}(a; \lambda, \tau) = \tau - \lambda c\tau \quad (112)$$

$$+ \int_{\mathbb{R}} g_1^\tau(x) \left\{ s(x) + \lambda \left[1 - \frac{g_0^\tau(x)}{g_1^\tau(x)} \right] r(x) \right\} dx. \quad (113)$$

Using Lemma 1, define

$$Q_\lambda(x) = \lambda \left[1 - \exp\left(\frac{\tau}{2} - x\right) \right] - 1. \quad (114)$$

Also define total net surplus from full second-phase effort by

$$C(h) = (1 - c)h. \quad (115)$$

The statewise Lagrangian values of the four extreme review actions are then

$$\mathcal{V}_0(x; \lambda) = 0, \quad (116)$$

$$\mathcal{V}_L(x; \lambda) = C(h) + Q_\lambda(x)r_L(h), \quad (117)$$

$$\mathcal{V}_H(x; \lambda) = C(h) + Q_\lambda(x)r_H(h), \quad (118)$$

$$\mathcal{V}_B(x; \lambda) = Q_\lambda(x)B. \quad (119)$$

The function Q_λ is strictly increasing whenever $\lambda > 0$:

$$Q'_\lambda(x) = \lambda \exp\left(\frac{\tau}{2} - x\right) > 0. \quad (120)$$

Consequently, the relative value of promising utility to the agent rises monotonically with review performance. Lemma 2 and equations (116)–(119) reduce the fixed-review problem to a pointwise comparison of four affine functions of the single increasing index $Q_\lambda(x)$.

The next section uses this comparison to prove that the optimal menu has at most four ordered regions: a bad stop, low-rent continuation, high-rent continuation, and a good stop. The multiplier λ and the associated cutoffs are then determined by pre-review obedience and complementary slackness.

5 The optimal review menu for a fixed review date

This section solves the principal's review-state assignment problem for a fixed review date $\tau \in (0, T)$. Let

$$h = T - \tau, \quad C(h) = (1 - c)h, \quad (121)$$

and suppress the dependence of $r_L(h)$ and $r_H(h)$ on h when no confusion can arise. We maintain strict feasibility of full effort in each phase:

$$B \left[2\Phi\left(\frac{\sqrt{t}}{2}\right) - 1 \right] > ct \quad \text{for } t \in \{\tau, h\}. \quad (122)$$

5.1 Pointwise comparison of the four review actions

For a multiplier $\lambda > 0$, the statewise Lagrangian values derived in the preceding section depend on the review state only through the increasing index

$$Q_\lambda(x) = \lambda \left[1 - \exp\left(\frac{\tau}{2} - x\right) \right] - 1. \quad (123)$$

It is therefore useful to compare the four actions as functions of a generic scalar q :

$$v_0(q) = 0, \quad (124)$$

$$v_L(q) = C(h) + qr_L, \quad (125)$$

$$v_H(q) = C(h) + qr_H, \quad (126)$$

$$v_B(q) = qB. \quad (127)$$

Define the two critical index values

$$q_0(h) = -\frac{C(h)}{r_L(h)} < 0 \quad (128)$$

and

$$q_2(h) = \frac{C(h)}{B - r_H(h)} > 0. \quad (129)$$

Define also

$$\underline{\lambda}(h) = \max \left\{ 0, 1 - \frac{C(h)}{r_L(h)} \right\}. \quad (130)$$

A multiplier supporting first-phase work must exceed $\underline{\lambda}(h)$; below this level the index never enters the low-rent continuation region on a set capable of supplying positive incentives.

Lemma 3 (Ordering of the four actions). *For any $q \in \mathbb{R}$, a maximizer of the four values in (127) can be chosen according to*

<i>Index value</i>	<i>Optimal action</i>	
$q < q_0(h)$	A_0 ,	(131)
$q_0(h) < q < 0$	A_L ,	
$0 < q < q_2(h)$	A_H ,	
$q > q_2(h)$	A_B .	

At $q = q_0(h)$, the actions A_0 and A_L are tied; at $q = 0$, the actions A_L and A_H are tied; and at $q = q_2(h)$, the actions A_H and A_B are tied.

Proof. For $q < 0$,

$$v_L(q) - v_H(q) = q[r_L - r_H] > 0, \quad (132)$$

so A_L dominates A_H . Moreover, $v_B(q) < 0 = v_0(q)$, so the only relevant comparison is between A_0 and A_L . Since

$$v_L(q) \geq v_0(q) \iff q \geq -\frac{C(h)}{r_L}, \quad (133)$$

the first two regions in (131) follow.

For $q > 0$, A_H dominates A_L , and $v_H(q) > 0 = v_0(q)$. It remains to compare A_H and A_B :

$$v_H(q) - v_B(q) = C(h) - q[B - r_H]. \quad (134)$$

Thus A_H is preferred when $q < q_2(h)$ and A_B is preferred when $q > q_2(h)$. The indifference statements follow directly. $\square$

5.2 The ordered cutoff menu

Because Q_λ is strictly increasing, Lemma 3 immediately yields an ordered review menu. Some upper regions may be absent because

$$\lim_{x \rightarrow \infty} Q_\lambda(x) = \lambda - 1. \quad (135)$$

Proposition 2 (At most four ordered review regions). *Fix $\tau \in (0, T)$ and suppose that first-phase work is implemented. Then an optimal review menu can be chosen to have at most four ordered regions:*

$$a_\lambda(x) = \begin{cases} A_0, & x < k_0(\lambda), \\ A_L, & k_0(\lambda) \leq x < k_1(\lambda), \\ A_H, & k_1(\lambda) \leq x < k_2(\lambda), \\ A_B, & x \geq k_2(\lambda), \end{cases} \quad (136)$$

where an absent upper region is represented by the corresponding cutoff being $+\infty$.

The bad-stop cutoff is always finite and satisfies

$$Q_\lambda(k_0(\lambda)) = -\frac{C(h)}{r_L(h)}. \quad (137)$$

Equivalently,

$$k_0(\lambda) = \frac{\tau}{2} - \log \left[1 - \frac{1}{\lambda} + \frac{C(h)}{\lambda r_L(h)} \right]. \quad (138)$$

The high-rent region is present if and only if $\lambda > 1$. In that case,

$$Q_\lambda(k_1(\lambda)) = 0 \quad (139)$$

and

$$k_1(\lambda) = \frac{\tau}{2} - \log \left(1 - \frac{1}{\lambda} \right). \quad (140)$$

If $\lambda \leq 1$, set $k_1(\lambda) = +\infty$.

The good-stop region is present if and only if

$$\lambda > 1 + \frac{C(h)}{B - r_H(h)}. \quad (141)$$

In that case,

$$Q_\lambda(k_2(\lambda)) = \frac{C(h)}{B - r_H(h)} \quad (142)$$

and

$$k_2(\lambda) = \frac{\tau}{2} - \log \left[1 - \frac{1}{\lambda} - \frac{C(h)}{\lambda [B - r_H(h)]} \right]. \quad (143)$$

Otherwise, set $k_2(\lambda) = +\infty$.

Whenever finite, the cutoffs are strictly ordered:

$$k_0(\lambda) < k_1(\lambda) < k_2(\lambda). \quad (144)$$

Consequently, the menu has the following possible forms:

Multiplier range	Review regions	
$\underline{\lambda}(h) < \lambda \leq 1$	$A_0 - A_L,$	
$1 < \lambda \leq 1 + \frac{C(h)}{B - r_H(h)}$	$A_0 - A_L - A_H,$	
$\lambda > 1 + \frac{C(h)}{B - r_H(h)}$	$A_0 - A_L - A_H - A_B.$	(145)

Proof. By Lemma 2, a pointwise maximizer may be selected from the four endpoint actions. Substituting $q = Q_\lambda(x)$ into Lemma 3 gives the ordered menu in (136), because Q_λ is strictly increasing.

The bad-stop cutoff solves (137). Solving this equation for k_0 gives (138). A menu capable of inducing first-phase work must satisfy

$$\lambda - 1 > -\frac{C(h)}{r_L(h)}, \quad (146)$$

for otherwise $Q_\lambda(x)$ never reaches the low-rent continuation region and the agent receives zero utility at every review state. Thus k_0 is finite.

The index Q_λ reaches zero at a finite state if and only if its upper limit in (135) is positive, which is equivalent to $\lambda > 1$. Solving $Q_\lambda(k_1) = 0$ gives (140).

Similarly, Q_λ reaches $q_2(h)$ at a finite state if and only if

$$\lambda - 1 > \frac{C(h)}{B - r_H(h)}, \quad (147)$$

which is exactly (141). Solving the corresponding indifference equation gives (143). Finally, the index thresholds satisfy

$$-\frac{C(h)}{r_L(h)} < 0 < \frac{C(h)}{B - r_H(h)}, \quad (148)$$

and Q_λ is strictly increasing, which proves (144). $\square$

Remark 2 (Economic interpretation of the four regions). *The bad stop assigns zero utility after sufficiently poor review performance. The low-rent continuation contract preserves second-phase production while providing the least costly positive reward for first-phase performance. Better review outcomes receive the high-rent continuation contract, which sacrifices some second-phase profit in order to strengthen first-phase incentives. For exceptionally high review outcomes, the principal may prefer the good stop: she pays the maximal prize B and abandons second-phase production because no work-inducing continuation contract can deliver as much utility to the agent.*

5.3 Pre-review obedience and the multiplier

For each finite cutoff, define its standardized value under first-phase work by

$$z_j(\lambda, \tau) = \frac{\tau - k_j(\lambda)}{\sqrt{\tau}}, \quad j \in \{0, 1, 2\}. \quad (149)$$

When a cutoff equals $+\infty$, set its standardized value equal to $-\infty$. Recall that

$$D(\tau, z) = \Phi(z) - \Phi(z - \sqrt{\tau}) \quad (150)$$

measures the increase in the probability of crossing the associated upper-tail cutoff when the agent works rather than shirks in the first phase.

Under the menu in Proposition 2, the agent's review-state utility can be written as a sum of utility increments:

$$r_\lambda(x) = r_L(h) \mathbf{1}\{x \geq k_0(\lambda)\} \quad (151)$$

$$+ [r_H(h) - r_L(h)] \mathbf{1}\{x \geq k_1(\lambda)\} \quad (152)$$

$$+ [B - r_H(h)] \mathbf{1}\{x \geq k_2(\lambda)\}. \quad (153)$$

Consequently, the incentive supplied by the menu is

$$I(\lambda; \tau) = r_L(h) D(\tau, z_0(\lambda, \tau)) \quad (154)$$

$$+ [r_H(h) - r_L(h)] D(\tau, z_1(\lambda, \tau)) \quad (155)$$

$$+ [B - r_H(h)] D(\tau, z_2(\lambda, \tau)). \quad (156)$$

The convention $D(\tau, -\infty) = 0$ automatically removes terms corresponding to absent regions.

Lemma 4 (Monotonicity of incentive supply). *The incentive supply $I(\lambda; \tau)$ is weakly increasing in λ . It is strictly increasing whenever a change in λ changes the selected review-state utility on a set of positive probability away from indifference points.*

Proof. Let $\lambda_2 > \lambda_1$, and let $(r_i(x), s_i(x))$ be pointwise Lagrangian maximizers for λ_i . Define

$$d_\tau(x) = 1 - \frac{g_0^\tau(x)}{g_1^\tau(x)}. \quad (157)$$

Pointwise optimality gives

$$s_2(x) + \lambda_2 d_\tau(x) r_2(x) \geq s_1(x) + \lambda_2 d_\tau(x) r_1(x), \quad (158)$$

$$s_1(x) + \lambda_1 d_\tau(x) r_1(x) \geq s_2(x) + \lambda_1 d_\tau(x) r_2(x). \quad (159)$$

Adding these inequalities yields

$$d_\tau(x) [r_2(x) - r_1(x)] \geq 0. \quad (160)$$

Multiplying by $g_1^\tau(x)$ and integrating gives

$$I(\lambda_2; \tau) - I(\lambda_1; \tau) = \int_{\mathbb{R}} g_1^\tau(x) d_\tau(x) [r_2(x) - r_1(x)] dx \geq 0. \quad (161)$$

The inequality is strict under the condition stated in the lemma. $\square$

Proposition 3 (Multiplier characterization). *Fix $\tau \in (0, T)$ and suppose (122) holds. Then an optimal solution to the fixed- τ problem conditional on inducing first-phase work can be chosen to have the ordered form in Proposition 2. Its multiplier $\lambda \geq 0$ satisfies complementary slackness:*

$$\lambda [I(\lambda; \tau) - c\tau] = 0. \quad (162)$$

Except for the knife-edge case in which the principal can provide first-phase incentives at zero marginal cost, the multiplier is strictly positive and pre-review obedience binds:

$$\boxed{I(\lambda; \tau) = c\tau.} \quad (163)$$

Any $\lambda > 0$ satisfying (163), together with the cutoffs in Proposition 2, implements an optimal fixed- τ review menu.

Proof. The fixed-review problem is a linear assignment problem with a single aggregate incentive constraint. Standard linear-programming duality yields a multiplier $\lambda \geq 0$ on pre-review obedience. For that multiplier, pointwise maximization of the Lagrangian selects the ordered menu in Proposition 2.

Complementary slackness gives (162). If the obedience constraint were slack at a regular optimum, then $\lambda = 0$. In that case the principal would choose the same surplus-maximizing review action at almost every state, so the agent's review-state utility would be constant and would provide no incentive for first-phase work. Thus, outside knife-edge indifference cases, $\lambda > 0$ and obedience binds.

For sufficiency, fix $\lambda > 0$ satisfying (163), and let a_λ be the associated pointwise Lagrangian maximizer. For any feasible menu a , pre-review obedience implies

$$\tau + \int s_a(x) g_1^\tau(x) dx \leq \mathcal{L}(a; \lambda, \tau). \quad (164)$$

Pointwise optimality of a_λ implies

$$\mathcal{L}(a; \lambda, \tau) \leq \mathcal{L}(a_\lambda; \lambda, \tau). \quad (165)$$

Because a_λ satisfies binding obedience,

$$\mathcal{L}(a_\lambda; \lambda, \tau) = \tau + \int s_{a_\lambda}(x) g_1^\tau(x) dx. \quad (166)$$

Hence no feasible menu gives the principal a larger payoff. $\square$

Remark 3 (Existence and possible nonuniqueness of the multiplier). *Outside the knife-edge case $C(h) = r_L(h)$, the cutoff formulas imply that $I(\lambda; \tau)$ is continuous on the relevant multiplier range. It approaches zero as λ decreases to $\underline{\lambda}(h)$, while Lemma 4 shows that it is weakly increasing. As $\lambda \rightarrow \infty$, the menu converges to a payment of B exactly when $X_\tau \geq \tau/2$, and hence*

$$I(\lambda; \tau) \longrightarrow B \left[2\Phi \left(\frac{\sqrt{\tau}}{2} \right) - 1 \right]. \quad (167)$$

This limit exceeds $c\tau$ by (122), so a solution to (163) exists. The solution is generically unique. If $C(h) = r_L(h)$, the bad stop and low-rent continuation action are tied at $\lambda = 0$; mixing between tied actions supplies any required incentive up to the corresponding threshold, so existence still obtains, although the supporting multiplier need not be unique.

5.4 When the upper review regions become active

Proposition 2 describes the menu in terms of the supporting multiplier. The monotonicity of incentive supply permits an equivalent characterization in terms of the primitive incentive requirement $c\tau$ for each fixed review date.

Proposition 4 (Boundaries between the two-, three-, and four-region menus). *Fix $\tau \in (0, T)$, set $h = T - \tau$, and suppose (122) holds. Define*

$$\zeta_0^2(\tau) = \frac{\sqrt{\tau}}{2} + \frac{1}{\sqrt{\tau}} \log \left(\frac{C(h)}{r_L(h)} \right) \quad (168)$$

and

$$\mathcal{I}_2(\tau) = r_L(h) D(\tau, \zeta_0^2(\tau)). \quad (169)$$

Then $\mathcal{I}_2(\tau)$ is the incentive supplied at the boundary $\lambda = 1$ at which the high-rent continuation region first appears. Consequently,

<i>First-phase incentive requirement</i>	<i>Upper review regions</i>	
$c\tau < \mathcal{I}_2(\tau)$	<i>no high-rent region,</i>	
$c\tau = \mathcal{I}_2(\tau)$	<i>boundary $\lambda = 1$,</i>	
$c\tau > \mathcal{I}_2(\tau)$	<i>a nonempty high-rent region.</i>	(170)

To characterize the good-stop boundary, define

$$q_2(h) = \frac{C(h)}{B - r_H(h)} \quad (171)$$

and

$$\zeta_0^3(\tau) = \frac{\sqrt{\tau}}{2} + \frac{1}{\sqrt{\tau}} \log \left[\frac{q_2(h) + C(h)/r_L(h)}{1 + q_2(h)} \right], \quad (172)$$

$$\zeta_1^3(\tau) = \frac{\sqrt{\tau}}{2} + \frac{1}{\sqrt{\tau}} \log \left[\frac{q_2(h)}{1 + q_2(h)} \right]. \quad (173)$$

Define

$$\mathcal{I}_3(\tau) = r_L(h) D(\tau, \zeta_0^3(\tau)) \quad (174)$$

$$+ [r_H(h) - r_L(h)] D(\tau, \zeta_1^3(\tau)). \quad (175)$$

Then $\mathcal{I}_3(\tau)$ is the incentive supplied at the boundary

$$\lambda = 1 + \frac{C(h)}{B - r_H(h)} \quad (176)$$

at which the good-stop region first appears. Generically, $\mathcal{I}_3(\tau) > \mathcal{I}_2(\tau)$, and the complete fixed- τ partition is

First-phase incentive requirement	Optimal review regions
$c\tau \leq \mathcal{I}_2(\tau)$	$A_0 - A_L,$
$\mathcal{I}_2(\tau) < c\tau \leq \mathcal{I}_3(\tau)$	$A_0 - A_L - A_H,$
$c\tau > \mathcal{I}_3(\tau)$	$A_0 - A_L - A_H - A_B.$

(177)

At either equality, the newly appearing region has zero probability.

Proof. At $\lambda = 1$, the high-rent cutoff is $+\infty$ and the menu uses only A_0 and A_L . Substituting $\lambda = 1$ into (138) and standardizing gives (168). Hence the incentive supplied at this boundary is exactly (169). Lemma 4 then gives (170).

The good-stop region first appears when

$$\lambda = 1 + q_2(h). \quad (178)$$

At this value, $k_2 = +\infty$. Substituting the boundary multiplier into (138) and (140), and then standardizing, gives (172) and (173). The incentive supplied at the boundary is therefore (175). The monotonicity of incentive supply implies that the good-stop region is absent below this level and present above it. Since the good-stop boundary multiplier strictly exceeds one, strict monotonicity at a regular solution gives $\mathcal{I}_3(\tau) > \mathcal{I}_2(\tau)$. $\square$

Remark 4 (Interpretation of the region boundaries). *The quantities $\mathcal{I}_2(\tau)$ and $\mathcal{I}_3(\tau)$ measure how much first-phase incentive can be supplied before the principal must use a more expensive reward instrument. Below $\mathcal{I}_2(\tau)$, termination and the minimum-rent continuation contract suffice. Above that boundary, the principal must reward better review outcomes with the high-rent continuation contract. Once the required incentive exceeds $\mathcal{I}_3(\tau)$, even the full range of continuation rents is insufficient at the margin, and the principal creates a top region in which she pays B and sacrifices second-phase production.*

5.5 The fixed-review value

Let $s_L(h) = C(h) - r_L(h)$ and $s_H(h) = C(h) - r_H(h)$. Conditional on first-phase work, the probabilities of the four regions are determined by the standardized cutoffs. The principal's value from the optimal fixed- τ menu is

$$V^R(\tau) = \tau \tag{179}$$

$$+ s_L(h) [\Phi(z_0(\lambda, \tau)) - \Phi(z_1(\lambda, \tau))] \tag{180}$$

$$+ s_H(h) [\Phi(z_1(\lambda, \tau)) - \Phi(z_2(\lambda, \tau))] \tag{181}$$

$$- B\Phi(z_2(\lambda, \tau)), \tag{182}$$

where λ satisfies (163) and $\Phi(-\infty) = 0$. Equation (182) applies to the two-, three-, and four-region cases under the cutoff conventions above.

6 Choosing the review date

We now allow the principal to choose the review date. For each $\tau \in (0, T)$, let $V^R(\tau)$ denote the value characterized in Section 5, conditional on inducing work throughout the first phase, and continue to write

$$h = T - \tau. \tag{183}$$

This section gives a dual representation that is convenient for varying τ , establishes the endpoint connection with the no-review benchmark, and reduces the choice of the review date to a one-dimensional optimization.

6.1 A dual representation

Let $Z \sim N(0, 1)$. Under first-phase work, the review state can be written as

$$X_\tau = \tau + \sqrt{\tau}Z. \tag{184}$$

Consequently,

$$\frac{g_0^\tau(X_\tau)}{g_1^\tau(X_\tau)} = \exp\left(-\frac{\tau}{2} - \sqrt{\tau}Z\right). \tag{185}$$

For $\lambda \geq 0$, define the random Lagrangian index

$$Q(\lambda, \tau, Z) = \lambda \left[1 - \exp\left(-\frac{\tau}{2} - \sqrt{\tau}Z\right) \right] - 1. \tag{186}$$

For a generic index value q and continuation horizon h , define

$$M(q, h) = \max \{0, C(h) + qr_L(h), C(h) + qr_H(h), qB\}, \tag{187}$$

where $C(h) = (1 - c)h$.

Proposition 5 (Dual representation of the review value). *For every $\tau \in (0, T)$ satisfying strict feasibility in both phases,*

$$\boxed{V^R(\tau) = \inf_{\lambda \geq 0} \Psi(\tau, \lambda)}, \tag{188}$$

where

$$\Psi(\tau, \lambda) = \tau - \lambda c\tau + \mathbb{E} [M(Q(\lambda, \tau, Z), T - \tau)]. \tag{189}$$

A minimizing multiplier is a multiplier supporting the optimal ordered menu. At a regular solution it is the unique $\lambda > 0$ satisfying

$$I(\lambda; \tau) = c\tau. \quad (190)$$

Proof. For a fixed multiplier, the Lagrangian in (113) is maximized pointwise over the review-action set. By Lemma 2, this pointwise maximization can be restricted to A_0, A_L, A_H, A_B . Their statewise values are exactly the four terms inside the maximum in (187), with $q = Q(\lambda, \tau, Z)$. Integrating under first-phase work gives (189). Strong duality for the fixed-review linear program then gives (188). The characterization of the minimizing multiplier follows from Proposition 3. $\square$

The representation in (188) is useful because the review date enters only through the length of the two phases, the Gaussian likelihood-ratio index, and the two extreme continuation rents.

6.2 Endpoint behavior and the no-review benchmark

The no-review benchmark arises at either endpoint of the review-date problem. At $\tau = 0$, there is no first phase and the principal simply uses the minimum-rent full-effort contract over the entire horizon. At $\tau = T$, there is no second phase and the review itself is the terminal performance signal.

Proposition 6 (Continuous extension to the endpoints). *Suppose full effort is strictly feasible for every phase length in $(0, T]$, and suppose the no-review full-effort contract is profitable. Then V^R is continuous on $(0, T)$ and admits a continuous extension to $[0, T]$ satisfying*

$$\boxed{V^R(0) = V^R(T) = V^{\text{NR}}(T; c, B).} \quad (191)$$

Consequently,

$$\sup_{\tau \in (0, T)} V^R(\tau) = \max_{\tau \in [0, T]} V^R(\tau) \geq V^{\text{NR}}(T; c, B). \quad (192)$$

The inequality in (192) reflects the principal's option to dispense with a genuine interior review. A review is strictly valuable if and only if

$$V^R(\tau) > V^{\text{NR}}(T; c, B) \quad (193)$$

for some $\tau \in (0, T)$.

Proof. For interior review dates, continuity follows from the implicit-function characterization of $z_L(h)$ and $z_H(h)$, continuity of the review-action set, and the maximum theorem applied to the fixed-review linear program. Changes in the number of review regions may create changes in the formula used to compute the value, but not jumps in the value itself.

Consider first $\tau \downarrow 0$. The largest principal payoff available from a second-phase action is $s_L(h) = C(h) - r_L(h)$ for all sufficiently small τ , because profitability of the no-review contract implies $s_L(T) = V^{\text{NR}}(T; c, B) > 0$. Hence

$$V^R(\tau) \leq \tau + s_L(T - \tau), \quad (194)$$

whose limit is $s_L(T) = V^{\text{NR}}(T; c, B)$.

For a matching lower bound, assign the low-rent continuation action A_L at every review state except on a sufficiently high upper tail, on which the principal assigns the good stop A_B . The utility increment on that tail is $B - r_L(h)$. Choose the hard upper-tail cutoff so that

$$[B - r_L(h)]D(\tau, z) = c\tau. \quad (195)$$

For sufficiently small τ , such a cutoff exists, and the work probability of the reward region converges to zero. The resulting payoff converges to $s_L(T)$. Therefore

$$\lim_{\tau \downarrow 0} V^R(\tau) = V^{\text{NR}}(T; c, B). \quad (196)$$

Now let $\tau \uparrow T$, so $h \downarrow 0$. A menu using only A_0 and A_B is exactly a bounded upper-tail bonus based on X_τ . Choosing its hard threshold to satisfy

$$BD(\tau, z) = c\tau \quad (197)$$

gives payoff $V^{\text{NR}}(\tau; c, B)$, which converges to $V^{\text{NR}}(T; c, B)$.

For the reverse inequality, the full-effort continuation frontier converges to the no-effort frontier. In particular,

$$C(h) \rightarrow 0, \quad r_L(h) \rightarrow 0, \quad r_H(h) \rightarrow B. \quad (198)$$

Thus the complete review-action set converges to

$$\{(r, -r) : r \in [0, B]\}. \quad (199)$$

The review signal converges to the terminal signal $N(T, T)$, and the fixed-review problem converges to the no-review bounded-payment problem. This proves

$$\lim_{\tau \uparrow T} V^R(\tau) = V^{\text{NR}}(T; c, B). \quad (200)$$

The remaining conclusions follow from continuity on the compactified interval $[0, T]$. $\square$

It is useful to define the review gain

$$G(\tau) = V^R(\tau) - V^{\text{NR}}(T; c, B). \quad (201)$$

Proposition 6 gives

$$G(0) = G(T) = 0. \quad (202)$$

Thus a valuable review appears as a strictly positive interior hump in the review-gain function. If $G(\tau) \leq 0$ throughout the interior, the principal optimally uses no review even though the fixed-review menu itself may have a rich cutoff structure.

6.3 One-sided endpoint derivatives

The endpoint equality in Proposition 6 makes one-sided derivatives especially informative. A negative right derivative at zero would favor a very early review, while a negative left derivative at T implies that moving the review slightly to the left of the terminal date strictly raises the principal's payoff. The two endpoints behave very differently.

Define the smallest payment cap that permits full effort over the entire horizon by

$$\underline{B}(T, c) = \frac{cT}{2\Phi(\sqrt{T}/2) - 1}. \quad (203)$$

For $B > \underline{B}(T, c)$, let z_T^L continue to denote the hard-threshold root in (28). Define

$$\ell_T = \frac{\phi(z_T^L - \sqrt{T})}{\phi(z_T^L)} = \exp\left(z_T^L \sqrt{T} - \frac{T}{2}\right) \in (0, 1) \quad (204)$$

and

$$\lambda_T^{\text{NR}} = \frac{1}{1 - \ell_T}. \quad (205)$$

The quantity λ_T^{NR} is the shadow value of the no-review incentive constraint.

Proposition 7 (Endpoint derivatives and tight payment caps). *Suppose full effort is strictly feasible over $[0, T]$ and the no-review full-effort contract is strictly profitable.*

1. *The review value has an infinitely negative right derivative at zero:*

$$\boxed{\lim_{\tau \downarrow 0} \frac{V^R(\tau) - V^{\text{NR}}(T; c, B)}{\tau} = -\infty.} \quad (206)$$

Consequently, there exists $\delta_0 > 0$ such that

$$V^R(\tau) < V^{\text{NR}}(T; c, B) \quad \text{for every } \tau \in (0, \delta_0). \quad (207)$$

2. *The left derivative at the terminal date equals the derivative of the no-review value with respect to its horizon:*

$$V_-^{R'}(T) = \frac{\partial V^{\text{NR}}(T; c, B)}{\partial T} \quad (208)$$

$$= 1 - \lambda_T^{\text{NR}} c + \lambda_T^{\text{NR}} \frac{B\phi(z_T^L - \sqrt{T})}{2\sqrt{T}} \quad (209)$$

$$= 1 - \frac{c - \frac{B\phi(z_T^L - \sqrt{T})}{2\sqrt{T}}}{1 - \exp(z_T^L \sqrt{T} - T/2)}. \quad (210)$$

In particular, a sufficient condition for a strictly valuable interior review is

$$\boxed{c > 1 - \ell_T + \frac{B\phi(z_T^L - \sqrt{T})}{2\sqrt{T}}.} \quad (211)$$

3. *The terminal derivative has the limiting behavior*

$$\lim_{B \downarrow B(T, c)} V_-^{R'}(T) = -\infty \quad (212)$$

and

$$\lim_{B \rightarrow \infty} V_-^{R'}(T) = 1 - c > 0. \quad (213)$$

Proof. We prove the three claims in turn.

For the first claim, write $h = T - \tau$ and use the low-rent continuation action $A_L(h)$ as a baseline. For all sufficiently small τ , this action uniquely maximizes the principal's post-review payoff. Compactness of the two continuation frontiers and strict profitability at $h = T$ imply that there is a constant $\kappa > 0$ such that every feasible review action (r, s) satisfies

$$s_L(h) - s \geq \kappa |r - r_L(h)|. \quad (214)$$

Let

$$L_\tau(x) = 1 - \frac{g_0^\tau(x)}{g_1^\tau(x)}. \quad (215)$$

Because a constant review utility provides no incentives, pre-review obedience can be written as

$$\mathbb{E}_1[(r(X_\tau) - r_L(h))L_\tau(X_\tau)] \geq c\tau. \quad (216)$$

Under work, $X_\tau = \tau + \sqrt{\tau}Z$ and

$$L_\tau(X_\tau) = 1 - \exp\left(-\frac{\tau}{2} - \sqrt{\tau}Z\right). \quad (217)$$

A standard Gaussian tail decomposition gives

$$\mathbb{E}_1[|r(X_\tau) - r_L(h)|] \geq K \frac{\sqrt{\tau}}{\sqrt{\log(1/\tau)}} \quad (218)$$

for some $K > 0$ and all sufficiently small τ . To see the order, split at

$$|Z| = \sqrt{8 \log(1/\tau)}. \quad (219)$$

On the central event, $|L_\tau|$ is of order $\sqrt{\tau \log(1/\tau)}$; under both work and shirking, the complementary Gaussian tails contribute $o(\tau)$. Since review utilities are bounded by B , (216) then implies (218).

Combining (214) and (218) yields

$$V^R(\tau) \leq \tau + s_L(T - \tau) - \kappa K \frac{\sqrt{\tau}}{\sqrt{\log(1/\tau)}}. \quad (220)$$

Since

$$\tau + s_L(T - \tau) = V^{\text{NR}}(T; c, B) + O(\tau), \quad (221)$$

(206) follows.

For the second claim, set $h = T - \tau$. As $h \downarrow 0$, the work-inducing continuation frontier differs from the no-effort frontier by a vertical surplus of only $(1 - c)h$. Moreover,

$$r_L(h) = O\left(\frac{\sqrt{h}}{\sqrt{\log(1/h)}}\right), \quad B - r_H(h) = O\left(\frac{\sqrt{h}}{\sqrt{\log(1/h)}}\right). \quad (222)$$

The continuation actions can therefore improve on the no-effort support function only on likelihood-index intervals whose widths are $O((\sqrt{h \log(1/h)}))$. Since the density of the review signal is bounded, their total incremental value is $o(h)$. Hence

$$V^R(T - h) = V^{\text{NR}}(T - h; c, B) + o(h). \quad (223)$$

This proves (208).

It remains to differentiate the no-review value. Implicit differentiation of

$$B \left[\Phi(z_T^L) - \Phi(z_T^L - \sqrt{T}) \right] = cT \quad (224)$$

gives

$$\frac{dz_T^L}{dT} = \frac{c/B - \phi(z_T^L - \sqrt{T})/(2\sqrt{T})}{\phi(z_T^L) - \phi(z_T^L - \sqrt{T})}. \quad (225)$$

Differentiating $V^{\text{NR}}(T; c, B) = T - B\Phi(z_T^L)$ and using (204) gives (209) and (210). If this derivative is negative, then $V^R(T - h) > V^R(T)$ for every sufficiently small $h > 0$, proving (211).

For the tight-cap limit, let $a = \sqrt{T}/2$. As $B \downarrow \underline{B}(T, c)$, the two no-review roots merge and

$$z_T^L \uparrow a, \quad \ell_T \uparrow 1. \quad (226)$$

The numerator in (210) converges to

$$c \left[1 - \frac{a\phi(a)}{2\Phi(a) - 1} \right], \quad (227)$$

which is strictly positive because

$$2\Phi(a) - 1 - a\phi(a) > 0 \quad \text{for every } a > 0. \quad (228)$$

Indeed, the expression on the left of (228) is zero at $a = 0$ and has derivative $(1 + a^2)\phi(a) > 0$. The denominator in (210) converges to zero from above, proving (212).

Finally, as $B \rightarrow \infty$, the hard root satisfies $z_T^L \rightarrow -\infty$, $\ell_T \rightarrow 0$, and $B\phi(z_T^L - \sqrt{T}) \rightarrow 0$. Equation (210) therefore yields (213). $\square$

The tight-cap conclusion can be stated directly in primitives.

Corollary 1 (Tight caps make an interior review strictly valuable). *Suppose*

$$c \leq \frac{2\Phi(\sqrt{T}/2) - 1}{\Phi(\sqrt{T}/2)}. \quad (229)$$

Equivalently, the no-review full-effort payoff is nonnegative at the feasibility boundary $B = \underline{B}(T, c)$. Then there exists $\varepsilon > 0$ such that, for every

$$B \in (\underline{B}(T, c), \underline{B}(T, c) + \varepsilon), \quad (230)$$

the no-review full-effort contract is profitable and

$$\boxed{\sup_{\tau \in (0, T)} V^R(\tau) > V^{\text{NR}}(T; c, B) \geq 0.} \quad (231)$$

Thus, whenever the cap is sufficiently close to the smallest cap that can induce full effort, a genuine interior review is strictly optimal.

Proof. At $B = \underline{B}(T, c)$, the no-review value is

$$T \left[1 - \frac{c\Phi(\sqrt{T}/2)}{2\Phi(\sqrt{T}/2) - 1} \right], \quad (232)$$

which is nonnegative exactly under (229). For caps just above the feasibility boundary, the hard-threshold implementation cost falls and the no-review payoff is nonnegative, strictly positive except possibly at the boundary itself. Proposition 7 implies that $V_-^R(T) < 0$ throughout a sufficiently small right neighborhood of $\underline{B}(T, c)$. Moving the review slightly to the left of T then strictly raises the principal's payoff. $\square$

Remark 5 (Probation, tight caps, and incentive capital). *Equation (206) has a sharp organizational interpretation. In this pure moral-hazard model, an arbitrarily short probationary phase has negative incentive value. A very early review is too noisy to distinguish work from shirking, yet bounded review rewards prevent the principal from magnifying its vanishing informational content at negligible cost. Any strictly optimal review must therefore be bounded away from date zero. Short probationary periods observed in practice must rest on motives absent from the model—for example, screening*

an unknown type, learning match quality, detecting technological failure, or preserving an option to abandon a bad project—rather than on the dynamic provision of effort incentives alone.

The derivative at $\tau = T$ reveals the opposite margin. Moving the review a small distance to the left creates a short continuation phase. That phase has little direct productive value, but it creates a menu of continuation rents. These rents can be assigned according to first-phase performance and thereby serve as incentive prizes. When the payment cap is tight, the no-review contract is close to the statistical limit of what bounded rewards can implement: the hard and easy incentive-compatible thresholds nearly coincide, the shadow value λ_T^{NR} becomes arbitrarily large, and a small amount of continuation incentive capital has a first-order value. Corollary 1 therefore strengthens the informal statement that a review is merely “likely” under a tight cap. Provided the no-review benchmark is nonnegative at the feasibility boundary, every sufficiently tight feasible cap makes an interior review strictly optimal.

With a loose cap, by contrast, the no-review contract can concentrate a large reward on a rare terminal outcome and induce effort with little rent. The shadow value of additional incentive instruments is then small, and (213) shows that the direct productive gain from lengthening the undivided project dominates near $\tau = T$. This does not rule out a more substantial interior review, but it eliminates the local terminal-date argument. The endpoint results thus expose two distinct requirements for a valuable review: the review must occur late enough to be informative, but early enough to leave a continuation phase capable of producing rents that can be recycled as incentives.

6.4 An envelope formula

The dual representation permits differentiation without differentiating the optimal multiplier. We first record the derivatives of the two extreme continuation rents. Let

$$D_z(h, z) = \phi(z) - \phi(z - \sqrt{h}), \quad (233)$$

$$D_h(h, z) = \frac{\phi(z - \sqrt{h})}{2\sqrt{h}}, \quad (234)$$

where ϕ is the standard normal density. Implicit differentiation of

$$BD(h, z_j(h)) = ch \quad (235)$$

gives

$$z'_j(h) = \frac{c/B - D_h(h, z_j(h))}{D_z(h, z_j(h))}, \quad j \in \{L, H\}, \quad (236)$$

and hence

$$r'_j(h) = B\phi(z_j(h))z'_j(h) - c. \quad (237)$$

For a statewise action selected by the maximum in (187), let $R(q, h)$ denote the associated agent utility:

$$R(q, h) = \begin{cases} 0, & A_0, \\ r_L(h), & A_L, \\ r_H(h), & A_H, \\ B, & A_B. \end{cases} \quad (238)$$

Also define the direct continuation-horizon derivative

$$H(q, h) = \begin{cases} 0, & A_0, \\ (1 - c) + qr'_L(h), & A_L, \\ (1 - c) + qr'_H(h), & A_H, \\ 0, & A_B. \end{cases} \quad (239)$$

At indifference points either adjacent value may be used; those points have zero probability under the Gaussian signal.

Proposition 8 (Derivative of the review value). *Suppose the supporting multiplier $\lambda(\tau)$ is locally unique and the solution is away from a knife-edge parameter value. Then V^R is differentiable at τ and*

$$\frac{dV^R(\tau)}{d\tau} = 1 - \lambda(\tau)c \quad (240)$$

$$+ \mathbb{E} [R(Q(\lambda, \tau, Z), h)Q_\tau(\lambda, \tau, Z) - H(Q(\lambda, \tau, Z), h)], \quad (241)$$

where $h = T - \tau$ and

$$Q_\tau(\lambda, \tau, Z) = \lambda \exp\left(-\frac{\tau}{2} - \sqrt{\tau}Z\right) \left(\frac{1}{2} + \frac{Z}{2\sqrt{\tau}}\right). \quad (242)$$

An interior optimal review date therefore satisfies

$$\frac{dV^R(\tau^*)}{d\tau} = 0. \quad (243)$$

At a regime transition, the corresponding condition uses the left and right derivatives.

Proof. Apply the envelope theorem to the dual problem in (188). Holding the minimizing multiplier fixed, differentiation of the first two terms in (189) gives $1 - \lambda c$. The derivative of the support function with respect to its first argument is the selected utility R . Because $h = T - \tau$, the direct horizon contribution enters with a minus sign and is $-H$. Differentiating (186) gives (242). Combining these terms yields (241). $\square$

Remark 6 (The review date balances information and incentive capital). *The first-order condition in (243) captures a particularly rich economic balancing act. Delaying the review lengthens the first phase. This raises expected output before the review, but it also raises the cost $c\tau$ of inducing first-phase work. These are the first two terms in (241).*

More importantly, delaying the review improves the statistical quality of the performance measure. Under work and shirking, the review-state means differ by τ while their common standard deviation is $\sqrt{\tau}$. Thus the standardized separation of the two distributions is $\sqrt{\tau}$, which increases with the review date. A later review makes it easier to direct high continuation utilities toward histories that are more indicative of work and to withhold those utilities after histories more indicative of shirking. In (241), this information effect operates through Q_τ and the selected review-state utility R .

The cost of waiting is that the continuation phase becomes shorter. The principal then has less productive continuation surplus, $C(h) = (1 - c)h$, to divide between herself and the agent. A longer continuation phase creates what may be called incentive capital: a family of work-inducing continuation contracts that generate real second-phase output while delivering different rents to the agent. By assigning these contracts according to the review outcome, the principal can recycle

rents that are needed for second-phase incentives into rewards for first-phase effort. As h falls, the productive value of this continuation menu shrinks, and strong first-phase rewards increasingly require sacrificing second-phase production, culminating in the good-stop action A_B . This continuation-horizon effect is represented by the term involving H in (241).

The optimal review date therefore does not merely trade an early noisy signal against a late precise signal. It balances the precision of the review against the amount of productive future contracting capacity left after the review. An early review preserves a valuable continuation phase but sorts the agent using a noisy measure of first-phase performance. A late review sorts performance more accurately but leaves little time in which continuation rents can be generated and recycled. The review is especially valuable when an interior date permits the principal to use future agency rents as prizes for past effort without giving up too much future production. This mechanism is purely contractual: because the Brownian technology is translation invariant, the review state does not change the productivity of second-phase effort.

6.5 Computing and comparing the optimal review

The preceding results reduce the endogenous-review problem to the following one-dimensional procedure.

1. For each $\tau \in (0, T)$, set $h = T - \tau$ and solve

$$BD(h, z) = ch \quad (244)$$

for $z_L(h)$ and $z_H(h)$.

2. Compute $r_L(h)$ and $r_H(h)$ from (69).
3. Solve the scalar obedience equation

$$I(\lambda; \tau) = c\tau \quad (245)$$

for the supporting multiplier.

4. Use Proposition 2 to recover the cutoffs and evaluate $V^R(\tau)$ from (182).
5. Maximize the resulting continuous function over $\tau \in [0, T]$.

The optimized principal value is

$$V^* = \max \left\{ 0, V^{\text{NR}}(T; c, B), \sup_{\tau \in (0, T)} V^R(\tau) \right\}. \quad (246)$$

Under the maintained assumption that the no-review full-effort contract is profitable, this simplifies to

$$V^* = \max_{\tau \in [0, T]} V^R(\tau). \quad (247)$$

There is no general ordering between an interior review and no review. The review can reduce agency rents by conditioning future opportunities on interim performance, but it may also destroy second-phase surplus through bad and good stops. Which effect dominates depends on (T, c, B) .

Remark 7 (Two numerical illustrations). *The following calculations solve the two continuation-root equations, the binding pre-review obedience equation, and the one-dimensional review-date problem. For*

$$T = 1, \quad c = 0.5, \quad B = 1.31, \quad (248)$$

the no-review value is

$$V^{\text{NR}} = 0.13379. \quad (249)$$

The optimal review is interior:

$$\tau^* = 0.47408, \quad V^R(\tau^*) = 0.20106. \quad (250)$$

Thus the review gain is approximately

$$0.06727. \quad (251)$$

The supporting multiplier is $\lambda = 2.60063$, and all four review regions are present. The standardized cutoffs are approximately

$$z_0 = 0.72305, \quad z_1 = -0.36064, \quad z_2 = -1.10160. \quad (252)$$

By contrast, for

$$T = 1, \quad c = 0.1, \quad B = 5, \quad (253)$$

the no-review value is

$$V^{\text{NR}} = 0.89387. \quad (254)$$

Numerical evaluation gives $V^R(\tau) < V^{\text{NR}}$ for every interior review date. The supremum is approached as the review moves to an endpoint, so the principal optimally uses no genuine review.

The examples show that the endpoint result in Proposition 6 is substantive rather than merely technical. The optimized review may be strictly valuable, and when it is valuable the full four-region menu can occur. For other parameters, the rich fixed-review contract is dominated by the simpler no-review benchmark. The next natural step is to map the regions of (T, c, B) in which an interior review is valuable and to study how the optimal review date and the number of active review regions vary with the primitives.

7 Relation to the literature

This project combines several ingredients that have each appeared separately in the contracting literature: bounded compensation, interim performance evaluation, termination following poor performance, and the design or timing of feedback. The closest antecedents help clarify both what is standard in the analysis and where the present model adds a distinct mechanism.

7.1 Bounded payments and likelihood-ratio bonuses

The static foundation is Jewitt et al. (2008), who study moral hazard with upper and lower bounds on compensation and show that optimal contracts place payments at the bounds over selected outcome regions. Under limited liability, their optimal contracts are naturally option-like. In a recent control setting, Jeong et al. (2026) derive likelihood-ratio-test payments when an agent chooses between controllers for a noisy linear dynamical system. These papers make clear that the upper-tail, zero-or- B bonus used in our one-phase problem is not by itself the source of novelty. Our contribution instead begins with what happens when such bounded contracts are embedded in a genuinely dynamic review problem.

7.2 Interim evaluation, continuation, and termination

Several papers study interim performance evaluation with sequential effort and limited liability. Chen and Chiu (2013) analyze a two-stage project in which compensation and second-stage effort can depend on an interim evaluation. Lukas (2023) studies interim evaluation when outcomes across periods are interdependent. These papers establish that an interim signal can affect both incentives and continuation decisions, but their stages are fixed and their signals and technologies differ from the Gaussian cumulative-output environment considered here.

Termination following an unfavorable interim outcome also has important precedents. In Ray (2007), evaluation creates value by sorting projects and ending those with low early returns. Chao et al. (2014) study incentives around a stage-gate decision in a setting that combines information and effort problems, while Wong (2023) endogenizes dynamic monitoring and employment decisions. The termination motive in our model is deliberately different. Because Brownian output is additive and the productivity of future effort is independent of the review state, continuation has the same technological value after every realization of X_τ . The bad stop is therefore purely incentive driven: it destroys otherwise valuable continuation opportunities in order to reward first-phase effort. At the other end of the performance distribution, the good stop pays B and forgoes second-phase effort because an extreme cash prize can be the cheapest way to supply sufficiently strong first-phase incentives.

7.3 Feedback design and review timing

The timing and disclosure of interim information have been studied in a variety of settings. Nafziger (2009) shows that the timing of information can have adverse incentive effects for a wealth-constrained agent. Ederer (2010) studies interim feedback in dynamic tournaments, where evaluation, motivation, and signal-jamming effects interact.

Most closely in spirit is the innovative work, Ely et al. (2025), who jointly design incentives and feedback in continuous time and obtain a silent phase followed by a transparent phase. Their principal controls what the agent learns about a coarse performance measure. If the agent *succeeds* before the end of the silent phase, the principal waits until the end of the phase to tell him – if he succeeds during the subsequent transparent phase, he is notified immediately. The principal’s objective in Ely et al. (2025) is to maximize the agent’s expected effort over the horizon, albeit without an explicit reason for doing so. By contrast, our principal receives the output produced by the agent net of compensation, which is the canonical agency objective. Here the review signal is public and Gaussian, and the principal chooses when to observe it and which bounded continuation opportunity to award in order to maximize this net-output payoff.

This distinction produces a different timing tradeoff. Waiting longer makes X_τ more informative about first-phase effort, but it shortens the continuation phase and thereby reduces the stock of future rents available for incentive provision. The review date balances statistical precision against what may be called *incentive capital*. The endpoint results sharpen that tradeoff: an arbitrarily short probationary phase has negative incentive value, whereas a sufficiently tight payment cap makes an interior review strictly optimal whenever the no-review full-effort benchmark is nonnegative at the feasibility boundary.

7.4 Positioning of the present results

To our knowledge, the cited literature does not combine an optimally timed public Gaussian review, phase-level binary effort, limited liability, a hard payment cap, and an endogenous menu of stopping and continuation contracts. The distinctive results are the following. First, unavoidable continuation

rents are recycled as prizes for first-phase effort. Second, strict MLRP and the linear continuation frontier generate an ordered menu with at most four regions: a bad stop, low-rent continuation, high-rent continuation, and a good stop. Third, both stopping regions are incentive instruments rather than responses to state-dependent continuation productivity. Finally, the review date is endogenous, and its endpoint derivatives yield sharp implications for probationary periods and for the role of tight versus loose payment caps.

The appropriate claim is therefore not that interim evaluation or threshold bonuses are new in isolation. Rather, the model identifies a new dynamic synthesis: future agency rents become scarce incentive resources, the principal allocates those resources through an ordered performance ladder, and the timing of the review determines both the quality of performance information and the quantity of rents available to reward it.

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